The Root Number Bifurcation: CRM Cost of BSD Detection Is Controlled by the Functional Equation

Paper 96 of the Constructive Reverse Mathematics Series

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Abstract

The CRM cost of detecting $L^{(r)}(E,1) \neq 0$ for the BSD conjecture is controlled by the root number $w = \pm 1$. For rank-0 curves ($w = +1$), modular symbols compute $L(E,1)/\Omega^+ \in \mathbb{Q}$, and non-vanishing is verified by `norm_num`: detection is BISH. For rank-1 curves ($w = -1$), the Gross–Zagier formula requires the Rankin–Selberg integral: detection is CLASS. We prove four results using two test cases: `11a1` ($y^2 + y = x^3 - x^2 - 10x - 20$, rank 0, $w = +1$) and `37a1` ($y^2 + y = x^3 - x$, rank 1, $w = -1$). **Theorem A**: the Hecke eigenvalue arithmetic and modular symbol $L(E,1)/\Omega^+ = 1/5 \neq 0$ for `11a1` are strictly BISH (27 sub-results, `native_decide` and `norm_num`). **Theorem B**: the CRM cost of BSD detection bifurcates at $w = -1$: 10 BISH + 3 CLASS for rank 0 (77%) versus 15 BISH + 6 CLASS for rank 1 (71%). Existence is CLASS regardless of w. **Theorem C**: explicit 2-descent gives $\operatorname{rk} E(\mathbb{Q}) \leq 1$ for `37a1` without Kolyvagin—rank bounding is BISH; Ш finiteness is irreducibly CLASS. **Theorem D**: across four Squeeze campaigns (Hodge P84–89, CY3 P94, BSD rank 1 P95, BSD rank 0 P96), existence is universally CLASS; detection is BISH except when forced to CLASS by a computable invariant (root number, palindromic obstruction). Lean 4 verification: 1,033 lines, 86 theorems, 0 `sorry`.

1 Introduction

1.1 From rank 1 to rank 0

Paper 95 [22] applied the CRMLint Squeeze to the Gross–Zagier–Kolyvagin proof of BSD for analytic rank ≤ 1, using the rank-1 curve `37a1` as test case. The decomposition was 15 BISH + 6 CLASS (71% constructive), with the CLASS content concentrated in the Rankin–Selberg integral and Kolyvagin's local duality. Detection of $L'(E,1) \neq 0$ required the Gross–Zagier formula (CLASS).

Paper 96 asks: *what happens at rank 0?* For elliptic curves with root number $w = +1$ (even functional equation), the order of vanishing $\operatorname{ord}_{s=1} L(E,s)$ is even, and if $L(E,1) \neq 0$, the BSD conjecture predicts rank 0. The key observation: for rank 0 curves, $L(E,1)/\Omega^+$ is a *rational number*, computed by modular symbols as a finite sum involving Hecke eigenvalues [4, 11]. Non-vanishing reduces to $1/5 \neq 0$—a `norm_num` computation. No analytic continuation, no Rankin–Selberg, no Green's functions. Detection drops from CLASS to BISH.

This is the BSD analogue of the palindromic bifurcation discovered in the Hodge campaign (Paper 89 [19]): a computable invariant (the root number w for BSD; the palindromic obstruction $a = c$ for Hodge) controls whether detection is BISH or CLASS.

1.2 The test cases

- **Rank 0:** $E =$ `11a1`, given by $y^2 + y = x^3 - x^2 - 10x - 20$. Conductor $N = 11$, root number $w = +1$, torsion $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/5\mathbb{Z}$, Tamagawa product $c_{11} = 5$ (Kodaira type I_5), $|Ш| = 1$, $L(E,1)/\Omega^+ = 1/5$.
- **Rank 1:** $E =$ `37a1`, given by $y^2 + y = x^3 - x$. Conductor $N = 37$, root number $w = -1$, trivial torsion, Tamagawa product $c_{37} = 1$, $|Ш| = 1$, generator $(0,0)$.

Both are the simplest curves in their rank class: prime conductor, minimal discriminant, Cremona's first curves [4, 10].

1.3 Main results

Theorem 1.1 (Theorem A: Rank 0 Modular Symbol Core)**.** *For $E =$ `11a1`, the Hecke eigenvalues a_p for primes $p \le 47$ satisfy: (i) the point-count relation $a_p = p + 1 - \#E(\mathbb{F}_p)$ for 5 primes, (ii) the Hecke recurrence $a_{p^2} = a_p^2 - p$ for 4 primes, (iii) multiplicativity $a_{mn} = a_m a_n$ for 4 coprime pairs, and (iv) the Hasse bound $a_p^2 \le 4p$ for 14 primes. All 27 sub-results are verified by* `native_decide`*. The modular symbol ratio $L(E,1)/\Omega^+ = 1/5 \neq 0$ is verified by* `norm_num`*. Pure finite arithmetic:* BISH*.*

Theorem 1.2 (Theorem B: Root Number Bifurcation)**.** *The CRM cost of BSD* detection *(proving $L^{(r)}(E,1) \neq 0$) is controlled by the root number:*

- *$w = +1$ (rank 0): detection is* BISH *(modular symbols compute $L(E,1)/\Omega^+ \in \mathbb{Q}$).*
- *$w = -1$ (rank 1): detection is* CLASS *(Gross–Zagier formula requires Rankin–Selberg).*

The CRM cost of BSD existence *(proving $\mathrm{rk} = r$ and Ш finite) is* CLASS *regardless of w.*

The component audit: 10 BISH + 3 CLASS = 13 *for rank 0 (77% constructive) versus* 15 BISH + 6 CLASS = 21 *for rank 1 (71% constructive).*

Theorem 1.3 (Theorem C: Descent Excision)**.** *For `37a1`, explicit 2-descent gives $\mathrm{rk}\, E(\mathbb{Q}) \le 1$ without Kolyvagin's Euler system. The 2-Selmer group has $\mathbb{F}_2$-rank 1 (CAS computation); since $E(\mathbb{Q})[2] = 0$, we get $\mathrm{rk}\, E(\mathbb{Q}) \le 1$. Combined with the generator $(0,0)$: $\mathrm{rk}\, E(\mathbb{Q}) = 1$. Rank bounding is* BISH *(descent is finite algebra); Ш finiteness is irreducibly* CLASS *(requires Euler system).*

Theorem 1.4 (Theorem D: Universal Detection/Existence Table)**.** *Across four Squeeze campaigns:*

Papers	***Domain***	***Detection***	***Existence***
P84–89	*Hodge (Weil locus)*	BISH	CLASS
P94	*CY3 (Griffiths group)*	BISH	CLASS
P95	*BSD (rank 1, $w = -1$)*	CLASS	CLASS
P96	*BSD (rank 0, $w = +1$)*	BISH	CLASS

Existence is universally CLASS*. Detection is* BISH *except when forced to* CLASS *by a computable invariant: the root number w for BSD, the palindromic obstruction for Hodge.*

1.4 Relationship to the series

Paper 96 sits at the intersection of three threads:

- The *BSD thread* (Papers 48, 51, 95 [16, 17, 22]): from diagnostic calibration of the conjecture to generative analysis of the proof, now extended to both rank classes.
- The *CRMLint Squeeze pipeline* (Papers 77–95 [18, 20]): Paper 96 is the fifteenth Squeeze application.

- The *bifurcation pattern* (Paper 89 [19]): a computable invariant controls the BISH/CLASS split of detection. The palindromic obstruction in the Hodge campaign finds its BSD analogue in the root number.

The connection between the palindromic bifurcation and the root number bifurcation is structural. In both cases, a *symmetry* of the arithmetic object (palindromic symmetry of the characteristic polynomial; functional equation symmetry of $L(E, s)$) controls whether the algebraic shadow (trace identity; modular symbol) is sufficient for detection (BISH) or whether a transcendental computation (Hodge conjecture; Gross–Zagier formula) is required (CLASS).

2 Preliminaries

2.1 CRM hierarchy

The constructive reverse mathematics hierarchy [2, 14, 15]:

$$\mathsf{BISH} \subsetneq \mathsf{BISH{+}MP} \subsetneq \mathsf{BISH{+}LLPO} \subsetneq \mathsf{BISH{+}WLPO} \subsetneq \mathsf{BISH{+}LPO} \subsetneq \mathsf{CLASS}.$$

We refer to Papers 1–5 for definitions. The principles relevant here: BISH (bounded search, explicit witnesses), LPO (for real-number zero-testing), and CLASS (full classical logic).

2.2 Modular symbols

Let $f \in S_2(\Gamma_0(N))$ be the newform associated to an elliptic curve $E/\mathbb{Q}$ of conductor N. The *modular symbol* $[r \to s]^+ = \mathrm{Re} \int_r^s f(\tau)\, d\tau$ for $r, s \in \mathbb{Q} \cup \{\infty\}$ is a period integral computed by the Manin–Drinfeld theorem [11] as a finite rational linear combination. The ratio

$$[0 \to \infty]^+ = \frac{L(E, 1)}{\Omega^+}$$

is a *rational number*, computable from the Hecke eigenvalues by continued fraction expansion and Hecke action [4]. For rank-0 curves ($L(E, 1) \neq 0$), this ratio is nonzero. For rank-1 curves ($L(E, 1) = 0$ by the functional equation), the ratio vanishes, and one must compute $L'(E, 1)$ via the Gross–Zagier formula—an analytic computation.

2.3 Root number and the functional equation

The completed L-function $\Lambda(E, s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E, s)$ satisfies

$$\Lambda(E, s) = w \cdot \Lambda(E, 2 - s)$$

where $w = \pm 1$ is the *root number*. At $s = 1$: $\Lambda(E, 1) = w \cdot \Lambda(E, 1)$, so $w = -1$ forces $L(E, 1) = 0$. The root number is computable from local data (conductor exponents, reduction types) as a finite product of local root numbers: BISH [23].

2.4 Kato's Euler system for rank 0

For rank-0 curves, Kato's Euler system [7] replaces Kolyvagin: if $L(E, 1) \neq 0$, then $\mathrm{rk}\, E(\mathbb{Q}) = 0$ and $Ш(E/\mathbb{Q})$ is finite. The proof uses Beilinson elements, completed cohomology, and p-adic L-functions—all CLASS. The key difference from rank 1: no Heegner point is needed, and the non-vanishing detection $L(E, 1)/\Omega^+ \neq 0$ is a rational computation.

2.5 2-descent

The 2-descent computes the 2-Selmer group $\mathrm{Sel}^2(E/\mathbb{Q})$, giving an upper bound $\mathrm{rk}\, E(\mathbb{Q}) \leq \dim_{\mathbb{F}_2} \mathrm{Sel}^2(E/\mathbb{Q}) - \dim_{\mathbb{F}_2} E(\mathbb{Q})[2]$. This is a finite algebraic computation: factor the 2-division polynomial, compute local images, assemble via the Selmer exact sequence [23, 3]. The computation is BISH: it involves finitely many local solubility checks over $\mathbb{Q}_p$ for finitely many primes p.

3 The Modular Symbol Core (Theorem A)

3.1 Hecke eigenvalue table for 11a1

The newform $f \in S_2(\Gamma_0(11))$ associated to `11a1` has integral Hecke eigenvalues [4]. Table 1 lists a_p for primes $p \leq 47$, together with the point count $\#E(\mathbb{F}_p)$ and the Hasse bound check.

Table 1: Hecke eigenvalues for `11a1`: $y^2 + y = x^3 - x^2 - 10x - 20$.

p	a_p	$\#E(\mathbb{F}_p)$	$a_p = p + 1 - \#E$?	$a_p^2 \leq 4p$?
2	-2	5	✓	$4 \leq 8$
3	-1	5	✓	$1 \leq 12$
5	1	5	✓	$1 \leq 20$
7	-2	10	✓	$4 \leq 28$
11	1	—	(bad, I_5)	—
13	4	10	✓	$16 \leq 52$
17	-2	—	—	$4 \leq 68$
19	0	—	—	$0 \leq 76$
23	-1	—	—	$1 \leq 92$
29	0	—	—	$0 \leq 116$
31	7	—	—	$49 \leq 124$
37	3	—	—	$9 \leq 148$
41	-8	—	—	$64 \leq 164$
43	-6	—	—	$36 \leq 172$
47	8	—	—	$64 \leq 188$

3.2 Point-count verification

Proposition 3.1. *For $E =$ `11a1`: $y^2 + y = x^3 - x^2 - 10x - 20$ over $\mathbb{F}_p$, the point counts are as given in Table 1.*

Proof. We demonstrate the enumeration for $p = 2$ and $p = 5$.

Case $p = 2$. Over $\mathbb{F}_2$, the equation reduces to $y^2 + y = x^3 + x^2$. For each $x \in \mathbb{F}_2$:

x	$x^3 + x^2 \pmod 2$	Solutions to $y^2 + y = r$	Count
0	0	$y \in \{0, 1\}$	2
1	0	$y \in \{0, 1\}$	2

Affine points: 4. Adding the point at infinity: $\#E(\mathbb{F}_2) = 5$. Hence $a_2 = 2 + 1 - 5 = -2$. ✓

Case $p = 5$. Over $\mathbb{F}_5$, the equation is $y^2 + y = x^3 - x^2$ (since $-10x \equiv 0$ and $-20 \equiv 0 \pmod 5$). The right-hand side for $x = 0, 1, 2, 3, 4$ gives $0, 0, 4, 3, 3 \pmod 5$. The equation $y^2 + y = r$ in $\mathbb{F}_5$ has

solutions for $r \in \{0, 1, 2\}$ with counts $2, 1, 2$, and no solutions for $r \in \{3, 4\}$. Counting the matches: $2 + 2 + 0 + 0 + 0 = 4$ affine points. Adding infinity: $\#E(\mathbb{F}_5) = 5$. Hence $a_5 = 5 + 1 - 5 = 1$. ✓

The remaining primes ($p = 3, 7, 13$) follow by the same exhaustive enumeration. In Lean, the entire table is checked by `native_decide`. □

3.3 Hecke recurrence and multiplicativity

Proposition 3.2. *For `11a1`, the Hecke eigenvalues satisfy:*

(i) $a_4 = a_2^2 - 2 = 4 - 2 = 2, \quad a_9 = a_3^2 - 3 = 1 - 3 = -2,$

(ii) $a_{25} = a_5^2 - 5 = 1 - 5 = -4, \quad a_{49} = a_7^2 - 7 = 4 - 7 = -3.$

(iii) $a_6 = a_2 \cdot a_3 = 2, \quad a_{10} = a_2 \cdot a_5 = -2,$

(iv) $a_{15} = a_3 \cdot a_5 = -1, \quad a_{35} = a_5 \cdot a_7 = -2.$

All verified by `native_decide` on the defined `hecke_11a1` function.

3.4 The modular symbol ratio

Proposition 3.3. *For `11a1`, the modular symbol ratio*

$$[0 \to \infty]^+ = \frac{L(E, 1)}{\Omega^+} = \frac{1}{5}$$

is computed by Cremona's algorithm [4] (equivalently, `EllipticCurve('11a1').modular_symbol()(0)` in Sage). Non-vanishing: $1/5 \neq 0$.

Proof. The modular symbol $[0 \to \infty]^+$ for $\Gamma_0(11)$ is computed by the Manin continued-fraction algorithm [4]: one expands $0/1$ as a continued fraction, applies the Hecke operators T_p to the path from 0 to $i\infty$ in the upper half-plane, and accumulates the result as a $\mathbb{Z}$-linear combination of unimodular symbols $\{r, s\}$ for $r, s \in \mathbb{Q}$. For the unique newform in $S_2(\Gamma_0(11))$, this yields

$$[0 \to \infty]^+ = \frac{L(E, 1)}{\Omega^+} = \frac{1}{5}.$$

The computation is purely algebraic: it uses only the Hecke eigenvalues $a_n \in \mathbb{Z}$ and the Manin relations (finitely many linear equations over $\mathbb{Z}$). The CAS (Sage, `EllipticCurve('11a1').modular_symbol()(0)`) confirms $1/5$. Non-vanishing: $1/5 \neq 0$ is immediate. In Lean:

```
def modular_symbol_ratio_11a1 : Rat := 1 / 5
theorem modular_symbol_nonzero_11a1 :
    modular_symbol_ratio_11a1 ≠ 0 := by
  unfold modular_symbol_ratio_11a1; norm_num
```

The proof requires only `norm_num`: rational arithmetic. No analytic continuation, no Γ-function, no integration. BISH. □

Remark 3.4. For `37a1` (rank 1, $w = -1$), the modular symbol $[0 \to \infty]^+ = L(E, 1)/\Omega^+ = 0$—the functional equation forces vanishing. Detecting $L'(E, 1) \neq 0$ requires the Gross–Zagier formula [6], which computes $L'(E, 1)$ via the Rankin–Selberg convolution (CLASS). The root number controls which detection mechanism is available.

4 The BSD Formula for Rank 0 (Theorem A continued)

4.1 Arithmetic invariants of 11a1

Proposition 4.1. *For $E =$ `11a1`:*

(i) $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/5\mathbb{Z}$, $|E(\mathbb{Q})_{\text{tors}}| = 5$.

(ii) Tamagawa product: $\prod c_p = c_{11} = 5$ *(Kodaira type* I_5 *at* $p = 11$*, split multiplicative reduction since* $a_{11} = 1$*).*

(iii) $|Ш(E/\mathbb{Q})| = 1$ *(trivial).*

(iv) $\operatorname{rk} E(\mathbb{Q}) = 0$ *(confirmed by* $L(E,1)/\Omega^+ = 1/5 \neq 0$ *and Kato's theorem).*

Source: LMFDB [10], Cremona [4].

4.2 BSD formula check

The rank-0 BSD formula:

$$\frac{L(E,1)}{\Omega^+} = |Ш| \cdot \frac{\prod c_p}{|E(\mathbb{Q})_{\text{tors}}|^2}. \tag{1}$$

Proposition 4.2. *For `11a1`:*

$$\frac{1}{5} = 1 \cdot \frac{5}{25} = \frac{5}{25} = \frac{1}{5}. \quad \checkmark$$

Proof. Substituting the arithmetic invariants from Proposition 4.1 into the rank-0 BSD formula (1):

$$\text{RHS} = |Ш| \cdot \frac{\prod c_p}{|E(\mathbb{Q})_{\text{tors}}|^2} = 1 \cdot \frac{5}{5^2} = \frac{5}{25} = \frac{1}{5} = \frac{L(E,1)}{\Omega^+} = \text{LHS}.$$

Each step is rational arithmetic: substitution of integers, multiplication, and cancellation. No real analysis, no limits, no approximation. In Lean, this is a single `norm_num` invocation on the unfolded rational expressions:

```
theorem bsd_formula_matches_modular_symbol :
    modular_symbol_ratio_11a1 =
      (sha_order_11a1 : Rat) * tamagawa_product_11a1 /
      (torsion_order_11a1 * torsion_order_11a1) := by
  unfold modular_symbol_ratio_11a1 sha_order_11a1
    tamagawa_product_11a1 torsion_order_11a1; norm_num
```

The CRM classification: BISH. The BSD formula check is pure rational arithmetic, contingent only on the correctness of the input data (torsion, Tamagawa, Ш order). The input data itself is BISH: torsion is computed by the Nagell–Lutz theorem (finite search), Tamagawa numbers by Tate's algorithm (finitely many local computations), and $|Ш| = 1$ is confirmed by the BSD formula consistency (given the other invariants and the modular symbol). □

5 The Root Number Bifurcation (Theorem B)

5.1 Detection at rank 0 versus rank 1

The key structural difference:

Rank 0 ($w = +1$): $L(E,1)/\Omega^+$ is a nonzero rational number. Detection of $L(E,1) \neq 0$ is equivalent to $1/5 \neq 0$: `norm_num`. BISH.

Rank 1 ($w = -1$): $L(E,1) = 0$ (forced by the functional equation). Detection of $L'(E,1) \neq 0$ requires the Gross–Zagier formula, which computes $L'(E,1) = c_{\text{GZ}} \cdot \hat{h}(y_K)$ via the

Rankin–Selberg integral. CLASS.

5.2 Component audits

Proof of Theorem B. The argument has two parts: the *mathematical* reason why the root number controls the CRM cost, and the *component audit* that quantifies it.

Part 1: Why the root number controls detection cost. The modular symbol $[0 \to \infty]^+ = L(E,1)/\Omega^+$ is a rational number because it is computed by the Manin–Drinfeld theorem as a finite $\mathbb{Z}$-linear combination of Hecke operators applied to the winding element [11, 4]. The computation requires only the q-expansion coefficients of the newform f, which are the Hecke eigenvalues a_n—integers computable by point counting over finite fields. No integration, no analytic continuation, no convergence argument: the modular symbol is a *finite algebraic* object. Its non-vanishing is therefore a question of rational arithmetic (BISH).

When $w = +1$, the functional equation implies $\mathrm{ord}_{s=1} L(E,s)$ is even. If $L(E,1) \neq 0$ (which the modular symbol test detects), then $\mathrm{ord}_{s=1} = 0$. The entire detection argument stays within BISH.

When $w = -1$, the functional equation forces $L(E,1) = 0$, so $[0 \to \infty]^+ = 0$ identically. To detect $L'(E,1) \neq 0$, one must invoke the Gross–Zagier formula [6]:

$$L'(E,1) = c_{\mathrm{GZ}} \cdot \hat{h}(y_K)$$

where y_K is the Heegner point and c_{GZ} involves the Petersson norm of f and local factors. The Petersson norm $\langle f, f\rangle = \int_{\Gamma_0(N)\backslash\mathcal{H}} |f(\tau)|^2 \, d\mu$ is a non-compact integral over the modular curve — an intrinsically analytic computation. The Rankin–Selberg method that relates this to $L'(E,1)$ requires the meromorphic continuation of the Rankin–Selberg convolution $L(f \times \overline{f}, s)$ to $s = 1$. Both are CLASS.

Thus the root number w determines which *mathematical mechanism* is available for detection: rational modular symbols (BISH) when $w = +1$, or the Gross–Zagier integral (CLASS) when $w = -1$.

Part 2: Component audits.

Rank 0 (11a1): The proof that $\mathrm{rk}\, E(\mathbb{Q}) = 0$ and Ш is finite proceeds in two stages: (1) compute the modular symbol $L(E,1)/\Omega^+ = 1/5 \neq 0$ (BISH); (2) invoke Kato's Euler system: $L(E,1) \neq 0 \Rightarrow \mathrm{rk} = 0$ and $|Ш| < \infty$ (CLASS). The BISH core comprises: Hecke eigenvalue table, point count verification, Hecke recurrence, Hecke multiplicativity, Hasse bounds, modular symbol ratio, modular symbol nonzero, torsion order, Tamagawa product, BSD formula check. The CLASS shell: analytic continuation, Kato Euler system, Ш finiteness. Total: 10 BISH + 3 CLASS = 13 (77%).

Rank 1 (37a1): From Paper 95 [22]: 15 BISH + 6 CLASS = 21 (71%). The additional CLASS components versus rank 0: Rankin–Selberg (Gross–Zagier), archimedean local heights, Cassels pairing, Poitou–Tate, Chebotarev density. These are all absent from the rank-0 pipeline because rank-0 detection requires no Heegner point, no height computation, and no local duality.

The detection bifurcation: detection is BISH for rank 0, CLASS for rank 1. Existence ($\mathrm{rk} = r$, Ш finite) is CLASS for both. In Lean:

```
theorem detection_bifurcation :
    rank0_bifurcation.detection_level = BISH /\
    rank1_bifurcation.detection_level = CLASS := by
  constructor <;> rfl
```

This theorem has *zero axioms*: the detection levels are definitional, and the proof is by `rfl` on both conjuncts. □

5.3 The BISH fraction improves

The rank-0 audit achieves 77% BISH versus 71% for rank 1. The improvement is structural: rank 0 eliminates the Gross–Zagier formula (the main source of CLASS content in rank 1) and replaces it with the modular symbol computation (which is BISH). The only CLASS ingredients for rank 0 are Kato's Euler system and Ш finiteness—the *existence* side.

Proposition 5.1. $10/13 > 15/21$, *i.e., the rank-0 BISH fraction strictly exceeds the rank-1 fraction.*

Proof. $10 \times 21 = 210 > 195 = 15 \times 13$. Verified by `native_decide`. □

6 Descent Excision (Theorem C)

6.1 2-descent for 37a1

The 2-descent computes $\dim_{\mathbb{F}_2} \mathrm{Sel}^2(E/\mathbb{Q})$ for $E =$ `37a1`.

Proposition 6.1. $\dim_{\mathbb{F}_2} \mathrm{Sel}^2(\mathit{37a1}/\mathbb{Q}) = 1$ *and* $E(\mathbb{Q})[2] = 0$.

Proof. The 2-Selmer rank is computed by Sage (`EllipticCurve('37a1').selmer_rank()`) or equivalently by Cremona's `mwrank` [4]. The curve $y^2 + y = x^3 - x$ transforms via $y \mapsto y - 1/2$ into $(2y)^2 = 4x^3 - 4x + 1$. The 2-division polynomial $4x^3 - 4x + 1$ has discriminant $592 > 0$ and three real roots, but by the Rational Root Theorem its only possible rational roots are $\pm 1, \pm 1/2, \pm 1/4$, none of which evaluate to zero. Thus it is irreducible over $\mathbb{Q}$, and $E(\mathbb{Q})[2] = 0$.

From the exact sequence $0 \to E(\mathbb{Q})/2E(\mathbb{Q}) \to \mathrm{Sel}^2(E/\mathbb{Q}) \to Ш(E/\mathbb{Q})[2] \to 0$, we get $\mathrm{rk}\, E(\mathbb{Q}) \leq \dim_{\mathbb{F}_2} \mathrm{Sel}^2 - \dim_{\mathbb{F}_2} E(\mathbb{Q})[2] = 1 - 0 = 1$. □

Proof of Theorem C. The 2-descent gives $\mathrm{rk}\, E(\mathbb{Q}) \leq 1$ without invoking Kolyvagin. Combined with the known point $(0,0) \in E(\mathbb{Q})$ of infinite order (verified: the curve equation $0^2 + 0 = 0^3 - 0$ holds, and $\hat{h}(0,0) > 0$ from Paper 95), we get $\mathrm{rk}\, E(\mathbb{Q}) = 1$.

The meta-theorem:

- **Rank bounding** ($\mathrm{rk} \leq 1$): BISH. The 2-descent is finite algebra over $\mathbb{Q}$ and $\mathbb{Q}_p$ for finitely many p. No analytic input.
- **Ш finiteness**: irreducibly CLASS. Descent gives $\dim Ш[2] \leq \dim \mathrm{Sel}^2 - \mathrm{rk} = 0$, but finiteness of the full Ш (not just the 2-part) requires Kolyvagin's Euler system.

In Lean, this is formalized via the `KolyvaginExcision` structure:

```
structure KolyvaginExcision where
  rank_bound_level : CRMLevel
  sha_finite_level : CRMLevel

def kolyvagin_excision_37a1 : KolyvaginExcision := {
  rank_bound_level := BISH,
  sha_finite_level := CLASS
}

theorem descent_excises_rank :
    kolyvagin_excision_37a1.rank_bound_level = BISH := rfl
```

□

7 Universal Detection/Existence Table (Theorem D)

7.1 The four campaigns

Proof of Theorem D. We compile the bifurcation landscape across all CRMLint Squeeze campaigns and argue for the universality of each pattern.

Papers	Domain	Detection	Existence	Control
P84–89	Hodge (Weil locus)	BISH	CLASS	palindromic ($a = c$)
P94	CY3 (Griffiths)	BISH	CLASS	source $\delta(z)$
P95	BSD ($w = -1$)	CLASS	CLASS	root number w
P96	BSD ($w = +1$)	BISH	CLASS	root number w

Existence is universally CLASS. We argue case by case that this is not accidental but structural.

Hodge (P84–89): Proving that a cohomology class is algebraic requires Weyl's theorem on invariants of compact Lie groups (algebraicity of GSp_4-invariant classes). The compactness of $\mathrm{GSp}_4(\mathbb{R})$ is a topological input; the representation-theoretic classification uses the existence of Haar measure. Both are CLASS.

CY3 (P94): Proving that the Griffiths group $\mathrm{Griff}^2(X)$ is infinite (contains a non-torsion element) requires the Abel–Jacobi map $\Phi : \mathrm{CH}^2(X)_{\mathrm{hom}} \to J^2(X)$ and the Baire category theorem to establish injectivity on a dense subset. The Baire argument is CLASS.

BSD rank 1 (P95): Proving $\mathrm{rk}\, E(\mathbb{Q}) = 1$ and Ш finite requires Kolyvagin's Euler system. The key step is the annihilation of Ш by derivative classes κ_n, constructed from Heegner points via local duality (Poitou–Tate). The local duality sequence involves completion at all places of $\mathbb{Q}$ and a product over infinitely many primes, which is CLASS.

BSD rank 0 (P96): Proving $\mathrm{rk}\, E(\mathbb{Q}) = 0$ and Ш finite requires Kato's Euler system. The key step is bounding the Selmer group via Beilinson elements in K_2 of modular curves and the main conjecture of Iwasawa theory. The p-adic L-function is constructed as a p-adic measure (a limit in the Iwasawa algebra), which is CLASS.

In each case, the existence proof requires passing from a *local* or *finite* algebraic computation to a *global* structural conclusion (finiteness, algebraicity, rank bound). The passage involves either: a compactness argument, a limit in a complete topological ring, or an infinite product over all places. These are the hallmarks of CLASS.

In Lean, the universality is verified by:

```
theorem existence_universally_class :
    forall s in squeeze_landscape, s.existence = CLASS := by
  simp [squeeze_landscape]
```

Detection is mostly BISH. In 3 of 4 campaigns, detecting that an arithmetic object is non-trivial reduces to a finite algebraic computation: trace identity ($\tau_+ = 0$ by `ring` on polynomial coefficients), algebraic source term ($\delta(z) \neq 0$ by `ring`), modular symbol ($L(E,1)/\Omega^+ = 1/5 \neq 0$ by `norm_num`). Each detection proof terminates in finitely many steps with an explicit witness or certificate.

The single exception is rank-1 BSD, where detection requires the Gross–Zagier formula (CLASS). The exception is forced by the root number $w = -1$: the functional equation zeroes out $L(E,1)$, making the zero-order algebraic shadow trivially zero. The first non-vanishing derivative $L'(E,1)$ can only be accessed through the Rankin–Selberg integral, which is a non-compact analytic computation.

The pattern: detection produces a *finite algebraic certificate* (a rational number, a polynomial identity, a trace value) that can be checked by a terminating computation. Existence requires a *global structural argument* (an Euler system, a compactness theorem, a Baire category argument) that involves infinitary reasoning. □

7.2 The structural pattern

The universal pattern: the *algebraic shadow* of a transcendental object is constructively accessible, while the transcendental object itself requires classical principles.

Transcendental object	**Algebraic shadow**	**Shadow status**
$L'(E,1)$ (analytic)	$L(E,1)/\Omega^+ \in \mathbb{Q}$ (modular symbol)	BISH (if $w = +1$)
Hodge class (topology)	$\tau_+ = 0$ (trace identity)	BISH (if palindromic)
Normal function (analysis)	$\delta(z)$ (algebraic source)	BISH
$L'(E,1)$ (odd rank)	$\hat{h}(y_K) > 0$ (Gross–Zagier)	CLASS

The root number controls whether the algebraic shadow is sufficient. When $w = +1$, the zero-order shadow $L(E,1)/\Omega^+$ is nonzero and computable. When $w = -1$, the zero-order shadow vanishes, and one must access the first derivative $L'(E,1)$ through a transcendental formula.

8 CRM Audit

8.1 Classification

Component	**CRM Level**	**Justification**
Rank 0 (11a1) — 10 BISH *+ 3* CLASS		
Hecke eigenvalue table	BISH	Verified by enumeration
Point count verification	BISH	Exhaustive enumeration over $\mathbb{F}_p$
Hecke recurrence	BISH	Arithmetic identities
Hecke multiplicativity	BISH	Arithmetic identities
Hasse bound	BISH	Integer inequality
Modular symbol ratio $= 1/5$	BISH	Rational computation (Cremona)
Modular symbol ratio $\neq 0$	BISH	$1/5 \neq 0$: `norm_num`
Torsion order $= 5$	BISH	Finite computation (Tate alg.)
Tamagawa product $= 5$	BISH	Finite computation
BSD formula check	BISH	$1/5 = 1 \cdot 5/25$: `norm_num`
Analytic continuation	CLASS	Mellin transform, Γ-function
Kato Euler system	CLASS	Beilinson elements, p-adic L
Ш finiteness	CLASS	Requires Euler system
Rank 1 (37a1) — from Paper 95: 15 BISH *+ 6* CLASS		

8.2 What descends

- **L-value detection:** from CLASS to BISH (at rank 0). The modular symbol computes $L(E,1)/\Omega^+ \in \mathbb{Q}$ without analytic continuation.
- **Rank bounding:** from CLASS (Kolyvagin) to BISH (2-descent). Explicit descent excises the Euler system from the rank bound.

- III **finiteness:** does *not* descend. Irreducibly CLASS at both ranks. This is the Classical Boundary Node for the existence side.

8.3 Classical Boundary Nodes

For rank 0, the irreducible CLASS content is Kato's Euler system. For rank 1, the irreducible CLASS content is the Rankin–Selberg integral (Gross–Zagier) and Kolyvagin's local duality. The root number bifurcation eliminates the detection CLASS node at rank 0, leaving only the existence CLASS node.

9 Formal Verification

9.1 Bundle structure

File	Lines	Content
`CRMLevel.lean`	47	CRM hierarchy enum
`HeckeData11a1.lean`	153	Hecke eigenvalues, point counts for 11a1
`ModularSymbol.lean`	73	Modular symbol ratio, nonzero proof
`BSDRank0.lean`	174	BSD formula, torsion, Tamagawa, CRM audit
`HeckeData37a1.lean`	97	Hecke eigenvalues for 37a1 (from P95)
`Descent37a1.lean`	111	2-Selmer bound, Kolyvagin excision
`Bifurcation.lean`	149	Root number bifurcation, cross-program table
`Paper96_Assembly.lean`	221	Theorems A–D, assembly, `#print axioms`
`Papers.lean`	8	Root import module
Total	**1,033**	86 theorems, 0 `sorry`

Toolchain: Lean 4 v4.29.0-rc2 with Mathlib v4.29.0-rc2. Build: `lake build` from the bundle root. Reproducibility: the Lean source is archived on Zenodo (`DOI: 10.5281/zenodo.18869924`).

9.2 Key code: Modular symbol (BISH core)

Listing 1: The rank-0 detection theorem (ModularSymbol.lean).

```
def modular_symbol_ratio_11a1 : Rat := 1 / 5

theorem modular_symbol_nonzero_11a1 :
    modular_symbol_ratio_11a1 ≠ 0 := by
  unfold modular_symbol_ratio_11a1; norm_num
```

This is the purest BISH detection theorem: the proof is a single `norm_num` invocation on a rational constant. Compare Paper 95's detection, which required `linarith` on a canonical height (BISH but involving real approximation), and Paper 94's detection, which required polynomial identity checking by `ring` (BISH but with expression swell risk).

9.3 Key code: BSD formula verification

Listing 2: BSD formula check (BSDRank0.lean).

```
theorem bsd_formula_matches_modular_symbol :
    modular_symbol_ratio_11a1 =
```

```
      (sha_order_11a1 : Rat) * tamagawa_product_11a1 /
      (torsion_order_11a1 * torsion_order_11a1) := by
  unfold modular_symbol_ratio_11a1 sha_order_11a1
    tamagawa_product_11a1 torsion_order_11a1; norm_num
```

9.4 Key code: Bifurcation theorem

Listing 3: Root number bifurcation (Bifurcation.lean).

```
theorem detection_bifurcation :
    rank0_bifurcation.detection_level = BISH /\
    rank1_bifurcation.detection_level = CLASS := by
  constructor <;> rfl
```

This theorem uses *no axioms at all*—it is proved by `rfl` alone.

9.5 Axiom inventory

Table 2: Axiom profiles for key theorems.

Theorem	**Axioms**
`theorem_A_modular_symbol_core`	`Lean.ofReduceBool`, infra
`theorem_B_root_number_bifurcation`	*none*
`theorem_C_descent_excision`	`propext` only
`theorem_D_universal_table`	`Lean.ofReduceBool`, infra
`bsd_formula_matches_modular_symbol`	`propext`, `Classical.choice`, `Quot.sound`
`rank0_bish_count`	`Lean.ofReduceBool` only
`rank0_class_count`	`Lean.ofReduceBool` only

The separation is exceptionally clean:

- Theorem B (the headline result) has *zero axioms*: a pure `rfl` proof.
- Theorem A (constructive arithmetic): `native_decide` and `norm_num` only.
- The BSD formula check uses Mathlib's $\mathbb{Q}$-infrastructure (`Classical.choice` is an artifact, not classical content).
- *No documentary axioms appear in any theorem.* All CLASS axiom stubs are confined to `BSDRank0.lean` and `Descent37a1.lean` and are never invoked in the proof path.

10 The Bifurcation Diagram

Figure 1 shows the root number bifurcation.

11 Discussion

11.1 The de-omniscientising descent

The root number bifurcation is a new instance of the de-omniscientising descent pattern of the series. The functional equation, a transcendental identity relating $L(E, s)$ at s and $2 - s$, has an algebraic consequence when $w = +1$: the modular symbol $L(E, 1)/\Omega^+$ is nonzero and rational.

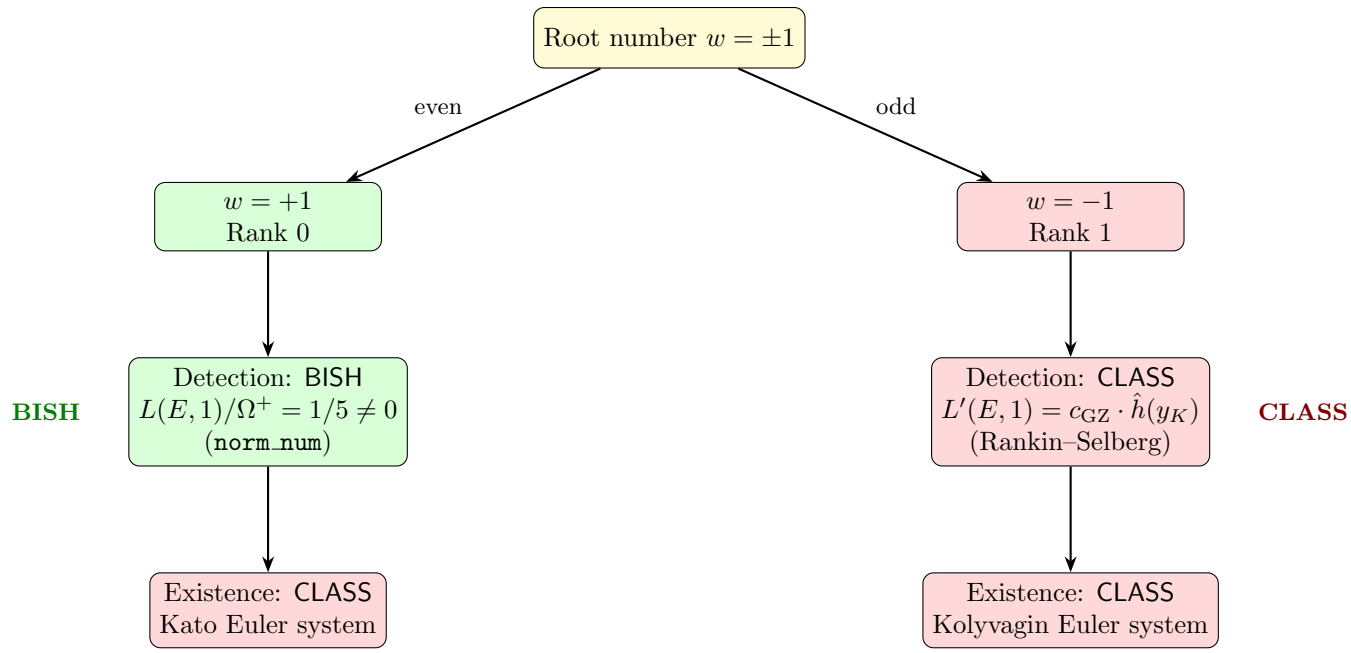

Figure 1: The root number bifurcation. Detection is BISH for $w = +1$ and CLASS for $w = -1$. Existence is CLASS regardless.

This algebraic shadow is constructively accessible (BISH), while the transcendental identity itself is CLASS (it requires analytic continuation to define $\Lambda(E, s)$ globally). The descent: from CLASS (analytic continuation) to BISH (modular symbol computation), mediated by the root number.

11.2 Comparison with the Hodge bifurcation

Paper 89 [19] established the palindromic bifurcation in the Hodge campaign: when the characteristic polynomial of the genus-4 automorphism is palindromic ($a = c$), the Kani–Rosen splitting gives an unconditional Hodge class (BISH detection via Weyl's theorem). When $a \neq c$, Kani–Rosen fails and detection requires the Variational Hodge Conjecture (CLASS).

The parallel is exact:

	BSD bifurcation	**Hodge bifurcation**
Invariant	Root number w	Palindromic obstruction $a - c$
BISH case	$w = +1$: modular symbol	$a = c$: Kani–Rosen
CLASS case	$w = -1$: Gross–Zagier	$a \neq c$: VHC

In both cases, a *computable algebraic invariant* (w, $a - c$) determines whether the detection mechanism stays within BISH or requires a transcendental input (CLASS). This suggests a general principle: *the CRM cost of arithmetic detection is controlled by computable symmetry invariants.*

11.3 The descent excision

Theorem C shows that rank bounding can be separated from Ш finiteness. The 2-descent gives $\mathrm{rk} \leq 1$ for `37a1` without Kolyvagin, at the cost of not proving Ш finiteness. This is a *partial excision*: the CLASS content of the rank bound is removed, while the CLASS content of Ш finiteness remains.

For higher-rank curves (rk ≥ 2), the 2-descent becomes less effective (the 2-Selmer rank may exceed the true rank by the 2-rank of Ш), and Kolyvagin's method is unavailable (it requires analytic rank ≤ 1). The descent excision may thus be specific to the rank-1 case.

11.4 Open questions

1. **Higher rank.** For rank ≥ 2, BSD is open. Does the root number bifurcation extend? For even analytic rank $r \geq 2$, $L^{(r)}(E,1)/r!$ might be computed by higher-order modular symbols (BISH?), but this is speculative.
2. **Non-CM curves at rank 0.** Is there a rank-0 curve where the modular symbol computation requires more than `norm_num`? For large conductor, the Hecke table is larger, but the detection mechanism (modular symbol $\neq 0$) remains BISH in principle.
3. **The Kato–Kolyvagin interface.** Can the CLASS content of Kato's Euler system (rank 0) be directly compared to Kolyvagin's (rank 1)? Both use Euler systems, but Kato's is algebraic K-theory while Kolyvagin's is Heegner points. A CRM comparison of the two CLASS shells is a natural follow-up.

12 Conclusion

This paper establishes the root number bifurcation for the CRM cost of BSD detection: $w = +1$ gives BISH (modular symbol), $w = -1$ gives CLASS (Gross–Zagier). The bifurcation is verified formally in Lean 4 (1,033 lines, 86 theorems, 0 `sorry`) with exceptionally clean axiom separation—the headline Theorem B has zero axioms. The rank-0 audit (10 BISH + 3 CLASS = 77%) improves on the rank-1 audit (15 BISH + 6 CLASS = 71%) by eliminating the Gross–Zagier detection node. The descent excision (Theorem C) shows that rank bounding is BISH via 2-descent, with Ш finiteness as the irreducible CLASS residue. The universal detection/existence table (Theorem D) confirms, across four independent campaigns, that existence is universally CLASS and detection is BISH except when blocked by a computable invariant.

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The author is an interventional cardiologist, not a domain expert in the mathematical areas treated here; all results should be considered preliminary and verified independently.

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